

Yi Fu (Summer)

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Experience

Software Engineer Intern - AI Systems | OpenOnion

Apr 2026 - Present

- Designed and built an LLM-based chat game using retrieval-augmented generation (RAG) and dynamic user modeling to deliver personalized dialogue, taking the product from spec to deployed MVP in under a week.
- Developing an LLM-powered web agent with interactive tool use, integrating internal APIs and tools to enable automated, multi-step task execution. Work contributed to a project that reached 1k+ GitHub stars.

Research Assistant | UNSW

Oct 2025 - Apr 2026

- Designed and trained transformer-based models for structured reasoning and planning tasks, improving performance on complex multi-step problem solving.
- Built LLM-based evaluation benchmarks using LLM-as-a-judge and newly designed metrics, achieving a 10% improvement over prior state-of-the-art; first-author paper under review.

Head Tutor & Course Administration | UNSW

Sept 2023 - Present

- Taught 80+ undergraduate and postgraduate students per term across software engineering topics, including version control (Git), Agile methodologies, full-stack and AI development.
- Designed supplementary course content and managed student forum (300-400 students) per term. Maintained a 6/6 teaching satisfaction rating across all 4 courses.

Student Researcher | UNSW

Feb 2023 - Jul 2025

- Designed a graph-based dataset generation pipeline with multiple features and representations for large-scale machine learning workflows, improving model accuracy by 10%. First-author publication in top ML conference AISTATS.
- Developed hybrid computational systems combining neural models with algorithmic solvers, reducing inference time of traditional solver by 20%. First-author paper under review.

UNSW Vertically Integrated (VIP) AI4Everyone Lab, Undergraduate

Feb 2022 - 2023

- Implemented a USAR Turtlebot3 in ROS2, Python and YOLOv5, improving accuracy in navigation and detection.

Projects

LLM-Based Multi-Agent Argument Simulator

- Built a multi-agent argument simulator in **Python** and **C#**, with role-based agents and hierarchical task decomposition.
- Integrated agent tool use, dynamic persona control, and RAG to enable adaptive, context-aware interactions.
- Implemented iterative reasoning loops with self-evaluation and reflection, improving response coherence, factual accuracy, and controllability.

AI Job Tracker Web App

- Built a full-stack job tracker using **Next.js**, **TypeScript**, **Prisma**, and **SQLite**, with AI extraction of role details from pasted job descriptions into structured JSON using the Claude and OpenAI APIs.
- Designed an AI-powered CV analysis workflow using PDF parsing and LLM-based requirement matching to turn unstructured application materials into structured gap analysis and tailored recommendations.

Visieyre VR Game

- Designed and developed first-person VR puzzle game using Unreal Engine (**Blueprint** & **C++**) with 3D rendering.
- Integrated computer vision-based pattern recognition system to detect and validate interactive user-drawn shapes.

Education

University of New South Wales (UNSW)

Sydney, NSW

Master of Philosophy in Computer Science and Engineering

Mar 2023 - Oct 2025

Bachelor of Science in Computer Science (UNSW Global Academic Award)

Sept 2020 - Feb 2023

Technologies

Programming & Web: Python, Java, C/C++/C#, JavaScript, TypeScript, React, Next.js, Node.js, REST APIs

AI & ML: PyTorch, RAG, Multi-agent workflows, Prompt engineering, Scikit-learn, LangChain, LangGraph, MCP

Databases & Backend Tools: SQL, PostgreSQL, MongoDB, JSON, Debugging, Testing

DevOps, Version Control & Platforms: Git, GitHub Actions, Docker, CI/CD, Linux, AWS, Azure, Google Colab